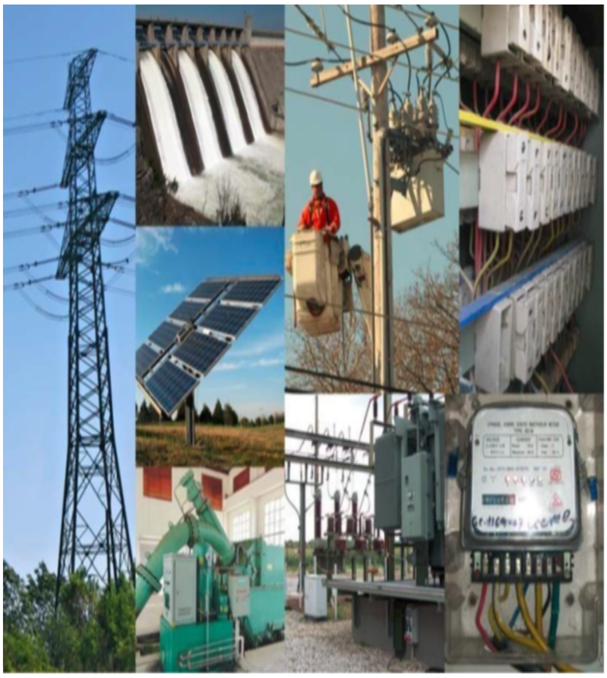
Qualification Pack



Distribution Network Helper

QP Code: PSS/Q2010

Version: 2.0

NSQF Level: 4

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri New Delhi-110021 || email:akbhagatin@yahoo.co.in

Contents

Qualification Pack

PSS/Q2010: Distribution Network Helper

Brief Job Description

Lineman Distribution constructs, operate and maintain overhead and underground Power Distribution system including distribution substation and cables including AB cables. This covers different pole structure, insulators, conductors, lightening arrestors, GO switch and distribution transformers etc. according to their specifications.

Personal Attributes

Physically and mentally able to safely perform essential functions of the job either with or without reasonable accommodations. The candidate should be able to climb ladders, scaffolds, poles and towers of various heights. The job holder should be able to cope with difficult work environment (such as heat, noise etc.). He/She must also have physical ability and stamina to stand for long working hours, besides mental ability to respond to emergency situation.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. PSS/N0105: Repair and Maintenance of Power Distribution Lines and components
2. PSS/N0106: Erection of Power Distribution Lines and Substations
3. PSS/N0107: Operation and Maintenance of an 11/0.433 KV Distribution Substation
4. PSS/N0108: Laying of Underground and AB Cables
5. PSS/N1336: Working effectively with others
6. PSS/N2001: Use basic health and safety practices for power related work

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Lineman
Country	India
NSQF Level	4

Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	8th Class (+ ITI Electrical 2 years) with 2 Years of experience OR 10th Class (+ ITI Electrical 1 year) with 1 Year of experience
Minimum Level of Education for Training in School	Same as Min.Ed for Job
Pre-Requisite License or Training	As per the guideline issued by Central Electricity Authority (CEA), Ministry of Power, Govt. of India/ respective State Govt. in this regard as amended from time to time
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/04/2024
NSQC Approval Date	31/03/2022
Version	2.0
Reference code on NQR	2022/POW/PSSC/05725
NQR Version	1.0

PSS/N0105: Repair and maintenance of Power Distribution Lines and components

Description

This unit covers the competencies required by technicians for repair and maintenance for Power Distribution Lines and components. This includes handling of tools and equipment for installation and maintenance and carrying out necessary repair and maintenance tasks in a safe, efficient and effective manner.

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

PC1. Work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. Adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations

PC3. Work according to laid down procedures and instructions

PC4. Ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location

PC5. Ensure work area is clean and safe from hazards before and after the job is completed

Prepare for repair and maintenance of power distribution lines

To be competent, the user/individual on the job must be able to:

PC6. access and survey area in accordance with established procedures

PC7. Assess and confirm condition of pole structure and components based on Distribution line standards

PC8. Perform load checks to identify imbalanced and overloaded circuits

PC9. identify hazards of trimming trees such as limits of approach, public safety and step and touch potential

PC10. Conduct site inspection for emergency cases following established procedures

PC11. identify various types of circuits

PC12. Identify and acquire correct tools, equipment and instruments required for Distribution line assessment and inspection

PC13. Ensure the tools and equipment is well maintained, calibrated and approved for use

PC14. Prepare and maintain the work area as per procedure or operation specification

PC15. prepare and maintain the work area as per procedure or operation specification

PC16. switch off, isolate, discharge and earth (side) line cables

PC17. confirm and/or obtain PTW/work permit (shut down) is taken to proceed to work from appropriate personnel in accordance with standard procedure

PC18. safely operate switchgears e.g. on/off, earth, etc.

Repair and maintenance of Power Distribution lines

To be competent, the user/individual on the job must be able to:

PC19. perform off-line overhead line maintenance procedure according to job specifications and requirements

PC20. perform off-line underground line maintenance procedure according to job specifications and requirements

PC21. perform stay wire assembly as per requirements and specifications, safely and efficiently

PC22. ensure lines are properly aligned by tightening appropriate nuts and bolts

PC23. ensure proper clearance of lowest conductor from ground

PC24. ensure guy insulators are of suitable capacity to the stay sets

PC25. select and use test equipment such as tong testers/clip-on meter, meggers and voltmeters to verify fault and integrity

PC26. sectionalize circuit to determine location of fault

PC27. isolate fault, damage or hazard and restore power to customers using equipment such as switches

PC28. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers

PC29. check work carried out by team members and ensure it is as per standard requirement

PC30. provide useful feedback regarding work matter to team members in a timely, polite and supportive manner

PC31. report trouble and required actions such as repairs or replacements, and estimated repair time to system authority

Carry out replacement activities as required

To be competent, the user/individual on the job must be able to:

PC32. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required

PC33. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging

PC34. replace components such as transformers, disconnects, conductors, poles, switches, elbows and terminations and insulators safely and as per company procedure

PC35. replace other line components due to damage or unsuitability as per standard procedure, where required

PC36. make connections and energize replaced underground cables, as per standard procedures where required

Post-repair and maintenance activities

To be competent, the user/individual on the job must be able to:

PC37. restore system to normal operating status by using switching procedures

PC38. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved

PC39. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities

PC40. refer unresolved job related problems to appropriate personnel for support

PC41. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. relevant health and safety requirements applicable in the work place

KU3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. how to engage with specialists for support in order to resolve incidents and service requests

KU6. importance of working in clean and safe environment practices and procedures

KU7. relevant people and their responsibilities within the work area

KU8. escalation matrix and procedures for reporting work and employment related issues

KU9. principles of electricity Principles: e.g. current, voltage, conductor size relation, series/parallel connections

KU10. common electricity terminology and correct interpretation of the same Terminology: e.g. Current, Voltage, Resistance, Inductance, Capacitance, Kilovolt ampere (kva), Kilowatt (kw), Kilowatt hour: (kwh)(unit of electric consumption), Power factor

KU11. specific terminology used in Distribution Line work Terminology: e.g. peak hours, peak load, load shedding, load transfer, Technical and commercial loss, maximum power,

KU12. elements of the power system Elements: e.g. generation, transmission, distribution, metering, equipment, etc

KU13. different types of material and accessories used in power Distribution Materials and accessories: e.g. Supports (Poles-Steel, Cement , Wooden), Conductors (Sizes, current carrying capacity), Conductor Accessories, Binding Tape, Binding Wire, P.G. Clamp, T Clamp etc. , switchgear panel, DT, Insulators (Pin, Disc, shackle, Guy etc.), Cross Arms, Stay sets, GO Switches etc. type of cross arms, etc.

KU14. tools and equipment used in testing, repair and maintenance Tools: e.g. Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), leakage current monitoring kit

KU15. specific health and safety precautions which must be taken when carrying out Distribution lines repair and maintenance work especially live line or equipment Precautions: e.g. loose dhotis, pajamas, key chain or watch chains should not be worn; shoes with projecting nails or other types of metal parts not to be used; do not start work unless circuit is in off condition and discharged, confirmation of line clear permit is taken on equipment, equipment or line is properly earthed

KU16. various types of circuits Types: e.g. C.T., P.T., A.C., D.C., Control, Series, Parallel, Neutral phase, Indication & Annunciation Circuits

KU17. troubleshooting and repair methods

KU18. fault indicators

KU19. overhead distribution system apparatus such as regulators and reclosers

KU20. overhead distribution system standards

KU21. access points such as vaults, open trenches and manholes

KU22. underground distribution system apparatus such as transformers, switching cubicles, distribution and junction boxes

KU23. co-existing underground utilities

KU24. causes of conductor damage Causes: Aeolian vibration, sway oscillation, galloping, unbalanced loading, over loading

KU25. classification of conductor and insulator damage including fretting, abrasion, fatigue breaks, tensile breaks

KU26. need for an authorized permit on 11 KV and above voltage line

KU27. hazards associated with carrying out power line maintenance and how they can be minimizedHazards: e.g. live wires, faulty insulation, voltage surges, faulty and damaged equipment and components, unsecure cables, unstable ladders, insects and reptiles, and scaffolding, etc

KU28. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively

KU29. importance of following good housekeeping and fire prevention procedures

KU30. importance of following job instructions and defined maintenance procedures

KU31. material preparation methods and techniques to be undertaken, prior to using for testing and maintenance activities

KU32. preparation of equipment for testing and repair activities

KU33. components of Distribution lines Line components: e.g. cross arm, insulator, line hardware, x-brace, armor rod, conductor, jumper, copper bond, arching horn, spacer, gang operated switch, drop out fuse, lightning arrester, etc.

KU34. procedures for handling Distribution line components with imperfections/defects that cannot be removed/repaired and how can they be minimized Imperfections/defects: e.g. Cross Arms (damaged cross arms, splitting or twisting, loose, broken, or missing nuts and braces, presence of insects), Insulators disc type (corroded pin, flashover, broken insulator, molds / moss or algae, hair crack), Insulator Synthetic polymer (broken rubber petticoat at hot end part, burned rubber petticoat at hot end part), Conductors (cut strand and loose conductor, loose vibration damper and spacer, low clearance (line to ground), Spot heating of connectors, other fittings and galvanized steel components (corroded bolts and nuts/steel pin, loose cotter key, dislocated steel pin, missing cotter / split pin), Ground wires and connectors (corroded earthwire, corroded / detached connector at jumper loop, corroded / cut ground lead, detached connector on ground lead and earthwire), Stay wires (rusted anchor rod, corroded)

KU35. problems and conditions which render electrical poles or towers in need of maintenance or replacement Problems and conditions: e.g. tower structure (corroded tower parts, loose or bent tower parts, eroded foundation), leaning pole, eroded pole, splitting, splitting or pulling of stay, twisting or raking, knots hole or birds nest, presence of insects, burned pole, excessive cracks, corroded poles, effects of lightning, etc.

KU36. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities

KU37. importance of reporting problems in a timely manner

KU38. methods and parameters to check quality of line components against required quality standards Methods: e.g. visual inspection, binoculars, measuring tape, use of instruments

KU39. principles and practices of electrical safety

KU40. standard procedures how to deal with electric shocks and electrocutions to rescue and minimize damage and harm

KU41. personal protective equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained PPE: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), zola, safety rope

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read/listen and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
GS2. fill up appropriate forms, activity logs, attendance sheets as per organizational format in English and/or local language
GS3. convey and share technical information clearly using appropriate language
GS4. check and clarify task-related information
GS5. liaise with appropriate authorities using correct protocol
GS6. communicate with people in respectful form and manner in line with organizational protocol
GS7. undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
GS8. identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder
GS9. use appropriate measuring techniques and units of measurement
GS10. use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
GS11. use metric systems of measurement
GS12. participate in on-the-job and other learning, training and development interventions and assessments
GS13. clarify task related information with appropriate personnel or technical adviser
GS14. seek to improve and modify own work practices
GS15. maintain current knowledge of application standards, legislation, codes of practice and product/process developments
GS16. identify problems with work planning, procedures, output and behavior and their implications
GS17. prioritize and plan for problem solving
GS18. communicate problems appropriately to others
GS19. identify sources of information and support for problem solving
GS20. seek assistance and support from other sources to solve problems
GS21. identify effective resolution techniques
GS22. select and apply resolution techniques
GS23. seek evidence for problem resolution

GS24. plan, prioritize and sequence work operations as per job requirements
GS25. organize and analyze information relevant to work
GS26. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
GS27. undertake and express new ideas and initiatives to others
GS28. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
GS29. participate in improvement procedures including process, quality and internal/external customer/supplier relationships
GS30. ones competencies in new and different situations and contexts to achieve more
GS31. exercise restraint while expressing dissent and during conflict situations
GS32. avoid and manage distractions to be disciplined at work
GS33. manage own time for achieving better results
GS34. work in a team in order to achieve better results
GS35. identify and clarify work roles within a team
GS36. communicate and cooperate with others in the team for better results
GS37. seek assistance from fellow team members

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working safely	**3**	**7**	**-**	**-**
PC1. Work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	1	2	-	-
PC2. Adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations	1	2	-	-
PC3. Work according to laid down procedures and instructions	1	1	-	-
PC4. Ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location	-	1	-	-
PC5. Ensure work area is clean and safe from hazards before and after the job is completed	-	1	-	-
Prepare for repair and maintenance of power distribution lines	**7**	**21**	**-**	**-**
PC6. access and survey area in accordance with established procedures	1	2	-	-
PC7. Assess and confirm condition of pole structure and components based on Distribution line standards	2	2	-	-
PC8. Perform load checks to identify imbalanced and overloaded circuits	-	2	-	-
PC9. identify hazards of trimming trees such as limits of approach, public safety and step and touch potential	-	2	-	-
PC10. Conduct site inspection for emergency cases following established procedures	1	2	-	-
PC11. identify various types of circuits	-	1	-	-
PC12. Identify and acquire correct tools, equipment and instruments required for Distribution line assessment and inspection	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Ensure the tools and equipment is well maintained, calibrated and approved for use	-	1	-	-
PC14. Prepare and maintain the work area as per procedure or operation specification	1	1	-	-
PC15. prepare and maintain the work area as per procedure or operation specification	1	1	-	-
PC16. switch off, isolate, discharge and earth (side) line cables	-	2	-	-
PC17. confirm and/or obtain PTW/work permit (shut down) is taken to proceed to work from appropriate personnel in accordance with standard procedure	1	2	-	-
PC18. safely operate switchgears e.g. on/off, earth, etc.	-	2	-	-
Repair and maintenance of Power Distribution lines	**10**	**26**	**-**	**-**
PC19. perform off-line overhead line maintenance procedure according to job specifications and requirements	2	2	-	-
PC20. perform off-line underground line maintenance procedure according to job specifications and requirements	2	2	-	-
PC21. perform stay wire assembly as per requirements and specifications, safely and efficiently	2	2	-	-
PC22. ensure lines are properly aligned by tightening appropriate nuts and bolts	-	2	-	-
PC23. ensure proper clearance of lowest conductor from ground	-	2	-	-
PC24. ensure guy insulators are of suitable capacity to the stay sets	-	2	-	-
PC25. select and use test equipment such as tong testers/clip-on meter, meggers and voltmeters to verify fault and integrity	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. sectionalize circuit to determine location of fault	-	2	-	-
PC27. isolate fault, damage or hazard and restore power to customers using equipment such as switches	-	2	-	-
PC28. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers	-	2	-	-
PC29. check work carried out by team members and ensure it is as per standard requirement	2	2	-	-
PC30. provide useful feedback regarding work matter to team members in a timely, polite and supportive manner	-	2	-	-
PC31. report trouble and required actions such as repairs or replacements, and estimated repair time to system authority	2	2	-	-
Carry out replacement activities as required	**4**	**11**	**-**	**-**
PC32. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required	2	2	-	-
PC33. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging	-	3	-	-
PC34. replace components such as transformers, disconnects, conductors, poles, switches, elbows and terminations and insulators safely and as per company procedure	1	2	-	-
PC35. replace other line components due to damage or unsuitability as per standard procedure, where required	1	2	-	-
PC36. make connections and energize replaced underground cables, as per standard procedures where required	-	2	-	-
Post-repair and maintenance activities	**1**	**10**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. restore system to normal operating status by using switching procedures	1	2	-	-
PC38. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	2	-	-
PC39. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	-	2	-	-
PC40. refer unresolved job related problems to appropriate personnel for support	-	2	-	-
PC41. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	2	-	-
NOS Total	**25**	**75**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0105
NOS Name	Repair and maintenance of Power Distribution Lines and components
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N0105: Repair and Maintenance of Power Distribution Lines and components

Description

This unit covers the competencies required by technicians for repair and maintenance for Power Distribution Lines and components. This includes handling of tools and equipment for installation and maintenance and carrying out necessary repair and maintenance tasks in a safe, efficient and effective manner.

Scope

The scope covers the following :

- This unit/task covers the following:
- 1. Working safely
- 2. Prepare for repair and maintenance of Power Distribution lines
- 3. Repair and maintenance of Power Distribution Lines
- 4. Carry out replacement activities as required
- 5. Post repair and maintenance activities

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

PC1. Work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. Adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations

PC3. Work according to laid down procedures and instructions

PC4. Ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location

PC5. Ensure work area is clean and safe from hazards before and after the job is completed

Prepare for repair and maintenance of power distribution lines

To be competent, the user/individual on the job must be able to:

PC6. access and survey area in accordance with established procedures

PC7. Assess and confirm condition of pole structure and components based on Distribution line standards

PC8. Perform load checks to identify imbalanced and overloaded circuits

PC9. identify hazards of trimming trees such as limits of approach, public safety and step and touch potential

PC10. Conduct site inspection for emergency cases following established procedures

PC11. identify various types of circuits

PC12. Identify and acquire correct tools, equipment and instruments required for Distribution line assessment and inspection

PC13. Ensure the tools and equipment is well maintained, calibrated and approved for use
PC14. Prepare and maintain the work area as per procedure or operation specification
PC15. prepare and maintain the work area as per procedure or operation specification
PC16. switch off, isolate, discharge and earth (side) line cables
PC17. confirm and/or obtain PTW/work permit (shut down) is taken to proceed to work from appropriate personnel in accordance with standard procedure
PC18. safely operate switchgears e.g. on/off, earth, etc.

Repair and maintenance of Power Distribution lines

To be competent, the user/individual on the job must be able to:

PC19. perform off-line overhead line maintenance procedure according to job specifications and requirements
PC20. perform off-line underground line maintenance procedure according to job specifications and requirements
PC21. perform stay wire assembly as per requirements and specifications, safely and efficiently
PC22. ensure lines are properly aligned by tightening appropriate nuts and bolts
PC23. ensure proper clearance of lowest conductor from ground
PC24. ensure guy insulators are of suitable capacity to the stay sets
PC25. select and use test equipment such as tong testers/clip-on meter, meggers and voltmeters to verify fault and integrity
PC26. sectionalize circuit to determine location of fault
PC27. isolate fault, damage or hazard and restore power to customers using equipment such as switches
PC28. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers
PC29. check work carried out by team members and ensure it is as per standard requirement
PC30. provide useful feedback regarding work matter to team members in a timely, polite and supportive manner
PC31. report trouble and required actions such as repairs or replacements, and estimated repair time to system authority

Carry out replacement activities as required

To be competent, the user/individual on the job must be able to:

PC32. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required
PC33. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging
PC34. replace components such as transformers, disconnects, conductors, poles, switches, elbows and terminations and insulators safely and as per company procedure
PC35. replace other line components due to damage or unsuitability as per standard procedure, where required
PC36. make connections and energize replaced underground cables, as per standard procedures where required

Post-repair and maintenance activities

To be competent, the user/individual on the job must be able to:

PC37. restore system to normal operating status by using switching procedures

PC38. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved

PC39. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities

PC40. refer unresolved job related problems to appropriate personnel for support

PC41. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. relevant health and safety requirements applicable in the work place

KU3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. how to engage with specialists for support in order to resolve incidents and service requests

KU6. importance of working in clean and safe environment practices and procedures

KU7. relevant people and their responsibilities within the work area

KU8. escalation matrix and procedures for reporting work and employment related issues

KU9. principles of electricity Principles: e.g. current, voltage, conductor size relation, series/parallel connections

KU10. common electricity terminology and correct interpretation of the same Terminology: e.g. Current, Voltage, Resistance, Inductance, Capacitance, Kilovolt ampere (kva), Kilowatt (kw), Kilowatt hour: (kwh)(unit of electric consumption), Power factor

KU11. specific terminology used in Distribution Line work Terminology: e.g. peak hours, peak load, load shedding, load transfer, Technical and commercial loss, maximum power,

KU12. elements of the power system Elements: e.g. generation, transmission, distribution, metering, equipment, etc

KU13. different types of material and accessories used in power Distribution Materials and accessories: e.g. Supports (Poles-Steel, Cement , Wooden), Conductors (Sizes, current carrying capacity), Conductor Accessories, Binding Tape, Binding Wire, P.G. Clamp, T Clamp etc. , switchgear panel, DT, Insulators (Pin, Disc, shackle, Guy etc.), Cross Arms, Stay sets, GO Switches etc. type of cross arms, etc.

KU14. tools and equipment used in testing, repair and maintenance Tools: e.g. Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), leakage current monitoring kit

KU15. specific health and safety precautions which must be taken when carrying out Distribution lines repair and maintenance work especially live line or equipment Precautions: e.g. loose dhotis, pajamas, key chain or watch chains should not be worn; shoes with projecting nails or other types of metal parts not to be used; do not start work unless circuit is in off condition and discharged, confirmation of line clear permit is taken on equipment, equipment or line is properly earthed

KU16. various types of circuits Types: e.g. C.T., P.T., A.C., D.C., Control, Series, Parallel, Neutral phase, Indication & Annunciation Circuits

KU17. troubleshooting and repair methods

KU18. fault indicators

KU19. overhead distribution system apparatus such as regulators and reclosers

KU20. overhead distribution system standards

KU21. access points such as vaults, open trenches and manholes

KU22. underground distribution system apparatus such as transformers, switching cubicles, distribution and junction boxes

KU23. co-existing underground utilities

KU24. causes of conductor damage Causes: Aeolian vibration, sway oscillation, galloping, unbalanced loading, over loading

KU25. classification of conductor and insulator damage including fretting, abrasion, fatigue breaks, tensile breaks

KU26. need for an authorized permit on 11 KV and above voltage line

KU27. hazards associated with carrying out power line maintenance and how they can be minimizedHazards: e.g. live wires, faulty insulation, voltage surges, faulty and damaged equipment and components, unsecure cables, unstable ladders, insects and reptiles, and scaffolding, etc

KU28. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively

KU29. importance of following good housekeeping and fire prevention procedures

KU30. importance of following job instructions and defined maintenance procedures

KU31. material preparation methods and techniques to be undertaken, prior to using for testing and maintenance activities

KU32. preparation of equipment for testing and repair activities

KU33. components of Distribution lines Line components: e.g. cross arm, insulator, line hardware, x-brace, armor rod, conductor, jumper, copper bond, arching horn, spacer, gang operated switch, drop out fuse, lightning arrester, etc.

KU34. procedures for handling Distribution line components with imperfections/defects that cannot be removed/repaired and how can they be minimized Imperfections/defects: e.g. Cross Arms (damaged cross arms, splitting or twisting, loose, broken, or missing nuts and braces, presence of insects), Insulators disc type (corroded pin, flashover, broken insulator, molds / moss or algae, hair crack), Insulator Synthetic polymer (broken rubber petticoat at hot end part, burned rubber petticoat at hot end part), Conductors (cut strand and loose conductor, loose vibration damper and spacer, low clearance (line to ground), Spot heating of connectors, other fittings and galvanized steel components (corroded bolts and nuts/steel pin, loose cotter key, dislocated steel pin, missing cotter / split pin), Ground wires and connectors (corroded earthwire, corroded / detached connector at jumper loop, corroded / cut ground lead, detached connector on ground lead and earthwire), Stay wires (rusted anchor rod, corroded)

KU35. problems and conditions which render electrical poles or towers in need of maintenance or replacement Problems and conditions: e.g. tower structure (corroded tower parts, loose or bent tower parts, eroded foundation), leaning pole, eroded pole, splitting, splitting or pulling of stay, twisting or raking, knots hole or birds nest, presence of insects, burned pole, excessive cracks, corroded poles, effects of lightning, etc.

KU36. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities

KU37. importance of reporting problems in a timely manner

KU38. methods and parameters to check quality of line components against required quality standards Methods: e.g. visual inspection, binoculars, measuring tape, use of instruments

KU39. principles and practices of electrical safety

KU40. standard procedures how to deal with electric shocks and electrocutions to rescue and minimize damage and harm

KU41. personal protective equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained PPE: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), zola, safety rope

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read/listen and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language

GS2. fill up appropriate forms, activity logs, attendance sheets as per organizational format in English and/or local language

GS3. convey and share technical information clearly using appropriate language

GS4. check and clarify task-related information

GS5. liaise with appropriate authorities using correct protocol

GS6. communicate with people in respectful form and manner in line with organizational protocol

GS7. undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages

GS8. identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder

GS9. use appropriate measuring techniques and units of measurement

GS10. use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity

GS11. use metric systems of measurement

GS12. participate in on-the-job and other learning, training and development interventions and assessments

GS13. clarify task related information with appropriate personnel or technical adviser

GS14. seek to improve and modify own work practices

GS15. maintain current knowledge of application standards, legislation, codes of practice and product/process developments

GS16. identify problems with work planning, procedures, output and behavior and their implications

GS17. prioritize and plan for problem solving

GS18. communicate problems appropriately to others

GS19. identify sources of information and support for problem solving

GS20. seek assistance and support from other sources to solve problems

GS21. identify effective resolution techniques

GS22. select and apply resolution techniques

GS23. seek evidence for problem resolution

GS24. plan, prioritize and sequence work operations as per job requirements

GS25. organize and analyze information relevant to work

GS26. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time

GS27. undertake and express new ideas and initiatives to others

GS28. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses

GS29. participate in improvement procedures including process, quality and internal/external customer/supplier relationships

GS30. ones competencies in new and different situations and contexts to achieve more

GS31. exercise restraint while expressing dissent and during conflict situations

GS32. avoid and manage distractions to be disciplined at work

GS33. manage own time for achieving better results

GS34. work in a team in order to achieve better results

GS35. identify and clarify work roles within a team

GS36. communicate and cooperate with others in the team for better results

GS37. seek assistance from fellow team members

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working safely	**3**	**7**	**-**	**-**
PC1. Work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	1	2	-	-
PC2. Adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations	1	2	-	-
PC3. Work according to laid down procedures and instructions	1	1	-	-
PC4. Ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location	-	1	-	-
PC5. Ensure work area is clean and safe from hazards before and after the job is completed	-	1	-	-
Prepare for repair and maintenance of power distribution lines	**7**	**21**	**-**	**-**
PC6. access and survey area in accordance with established procedures	1	2	-	-
PC7. Assess and confirm condition of pole structure and components based on Distribution line standards	2	2	-	-
PC8. Perform load checks to identify imbalanced and overloaded circuits	-	2	-	-
PC9. identify hazards of trimming trees such as limits of approach, public safety and step and touch potential	-	2	-	-
PC10. Conduct site inspection for emergency cases following established procedures	1	2	-	-
PC11. identify various types of circuits	-	1	-	-
PC12. Identify and acquire correct tools, equipment and instruments required for Distribution line assessment and inspection	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Ensure the tools and equipment is well maintained, calibrated and approved for use	-	1	-	-
PC14. Prepare and maintain the work area as per procedure or operation specification	1	1	-	-
PC15. prepare and maintain the work area as per procedure or operation specification	1	1	-	-
PC16. switch off, isolate, discharge and earth (side) line cables	-	2	-	-
PC17. confirm and/or obtain PTW/work permit (shut down) is taken to proceed to work from appropriate personnel in accordance with standard procedure	1	2	-	-
PC18. safely operate switchgears e.g. on/off, earth, etc.	-	2	-	-
Repair and maintenance of Power Distribution lines	**10**	**26**	**-**	**-**
PC19. perform off-line overhead line maintenance procedure according to job specifications and requirements	2	2	-	-
PC20. perform off-line underground line maintenance procedure according to job specifications and requirements	2	2	-	-
PC21. perform stay wire assembly as per requirements and specifications, safely and efficiently	2	2	-	-
PC22. ensure lines are properly aligned by tightening appropriate nuts and bolts	-	2	-	-
PC23. ensure proper clearance of lowest conductor from ground	-	2	-	-
PC24. ensure guy insulators are of suitable capacity to the stay sets	-	2	-	-
PC25. select and use test equipment such as tong testers/clip-on meter, meggers and voltmeters to verify fault and integrity	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. sectionalize circuit to determine location of fault	-	2	-	-
PC27. isolate fault, damage or hazard and restore power to customers using equipment such as switches	-	2	-	-
PC28. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers	-	2	-	-
PC29. check work carried out by team members and ensure it is as per standard requirement	2	2	-	-
PC30. provide useful feedback regarding work matter to team members in a timely, polite and supportive manner	-	2	-	-
PC31. report trouble and required actions such as repairs or replacements, and estimated repair time to system authority	2	2	-	-
Carry out replacement activities as required	**4**	**11**	**-**	**-**
PC32. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required	2	2	-	-
PC33. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging	-	3	-	-
PC34. replace components such as transformers, disconnects, conductors, poles, switches, elbows and terminations and insulators safely and as per company procedure	1	2	-	-
PC35. replace other line components due to damage or unsuitability as per standard procedure, where required	1	2	-	-
PC36. make connections and energize replaced underground cables, as per standard procedures where required	-	2	-	-
Post-repair and maintenance activities	**1**	**10**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. restore system to normal operating status by using switching procedures	1	2	-	-
PC38. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	2	-	-
PC39. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	-	2	-	-
PC40. refer unresolved job related problems to appropriate personnel for support	-	2	-	-
PC41. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	2	-	-
NOS Total	**25**	**75**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0105
NOS Name	Repair and Maintenance of Power Distribution Lines and components
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	2.0
Version	1.0
Next Review Date	NA

PSS/N0106: Erection of Power Distribution Lines and Substations

Description

This unit covers the competencies required technicians to erect and commissioningfor Power Distribution Lines. This includes working with the crew to install poles,dismantle poles and lay wiring, handling of tools and equipment for installation andcarrying out necessary tasks in a safe, efficient and effective manner.The candidate will be expected to perform mostly independently with little or nosupervision and as per job specifications.

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations

PC3. work following laid down procedures and instructions

PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location

PC5. ensure work area is clean and safe from hazards before and after the job is completed

Prepare for erection of Power Distribution Linea and Substation

To be competent, the user/individual on the job must be able to:

PC6. identify job requirements for specific operations as per instructions given from valid sources valid sources: job instruction sheet/job card; work drawings; supervisor/in- charge

PC7. brief team members as per requirement, agree and clarify role and job requirements and specifications

PC8. ensure equipment and tools required for distribution installation work are identified, acquired, calibrated, suitable and approved for use

PC9. carry out survey of proposed route of line

PC10. identify, estimate and acquire correct materials required for the installation work

PC11. ensure loading and unloading operations for pole parts in a safe and efficient manner

Erection of Power Distribution Lines and Sub- Stations

To be competent, the user/individual on the job must be able to:

PC12. determine pole location(s) as per approved procedures

PC13. ensure excavation operations are carried out with the help of ground crew for pole setting template, as per requirement and specifications, in a safe and efficient manner

PC14. perform pole erection procedures as per requirements and specifications, in a safe and efficient manner

PC15. install grounding for pole installation where required and cross arm fixing to the pole before erection

PC16. ensure pit filling and concreting is done as per requirement, as correct procedures

PC17. follow applicable construction standards e.g. rec construction standards, for carrying out the erection procedures

PC18. install pole guys/ stays and anchors as required, as per standard procedure

PC19. perform stay wire assembly as per requirements and specifications, safely and efficiently

PC20. install travelers on poles or insulators

PC21. temporarily run conductor/rope through travelers to reduce friction when sagging

PC22. attach pulling equipment to conductor/rope

PC23. set up and operate stringing equipment when using tension stringing method, derrick method.

PC24. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging

PC25. transfer conductor from travelers to insulators

PC26. secure conductor using clamps or ties

PC27. carry out erection of poles mounted distribution sub-station as per rec constructional standard f2

PC28. ensure maintenance of necessary clearances for execution of poles mounted distribution sub-station

PC29. carry out earthing arrangements for pole mounted distribution sub-station as per rec constructional standard f10

PC30. ensure 3 separate earthing with each for lightining arrestor, transformer body with metal parts and neutral

PC31. ensure proper rating of lt mains cable & fuses (ht & lt) during pole mounting sub-station installation

Commissioning

To be competent, the user/individual on the job must be able to:

PC32. thoroughly check the line for clearances

PC33. check guarding and stays for correctness and suitability

PC34. install warning devices and signage

PC35. inspect the pole and related components to check if it is as per specification and without defects

PC36. ensure that before commissioning the distribution transformer, all earthing connections are carried out and earth resistance is maintained within limits

PC37. check breather, oil level in the conservator, ht & lt side jumper connections

PC38. carry out pre-commissioning test of distribution transformer viz. ir test, continuity test etc

Post Erection activities

To be competent, the user/individual on the job must be able to:

PC39. shut down and store equipment to a safe condition on completion of the activities

PC40. leave the work area in a safe and tidy condition on completion of the erection activities

PC41. refer unresolved job related problems to appropriate personnel for support

PC42. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. relevant health and safety requirements applicable in the work place

KU3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. how to engage with specialists for support in order to resolve incidents and service requests

KU6. importance of working in clean and safe environment practices and procedures

KU7. relevant people and their responsibilities within the work area

KU8. escalation matrix and procedures for reporting work and employment related issues

KU9. standards and specifications for stringing of overhead line including accessories

KU10. basic elements of electricity Elements: e.g. current, voltage, conductor relationships, KWI

KU11. kinds of tools and equipment used in erection of poles and towers Tools and equipment: e.g. sling, drilling machine, climbing gear, digging tools, wrench set, hammer, tools bag, block and tackle with rope, ratchet set, hand winch, compression tool, hydraulic cutter, boom truck, pulleys (force pulley with sling), come along clamp, max puller, tension meter

KU12. Tower parts and accessories Parts and accessories: e.g. insulator, machine bolts, suspension clamps, strain clamp, overhead earth wires, cross-arms and braces, conductors and accessories, bolts and nuts, plates and back plates, grounding cables)

KU13. specific health and safety precautions which must be taken when carrying out pole erection procedures Safety requirements: e.g. poles securely fastened, warning devices are installed

KU14. hazards associated with carrying out pole erection processes and how they can be minimized Hazards: e.g. blockages and obstructions, live wires and equipment,unsecured ladders, etc.

KU15. circuit breaker, Isolators, their purpose and functioning

KU16. LT/HT transmission system and its components

KU17. importance of following job instructions and defined procedures for tower/pole erection

KU18. methods and procedures for pole erection Procedures: e.g. derrick method, gin pole method, knot tying / splicing, pole alignment

KU19. stringing method and procedures to be used Procedures: e.g. conductor / overhead (ACSR stringing) conductor mid span joints, install stay set (guys and anchor), pole dressing (muffler coping, painting and anti-climbing devices), fitting of vibration damper and arching horn

KU20. importance of correct sagging and following correct sag-tension

KU21. problems that can occur with the erection operations, and how these can be overcome

KU22. importance of leaving the work area and equipment in a safe and clean condition on completion of the erection activities

KU23. importance of reporting problems in a timely manner

KU24. methods and parameters to check quality of the components against required quality standards

KU25. importance and procedures to keep record of the job including data logging, chart recording of various activities and data points like tolerance levels, etc

KU26. importance of tools and equipment to be kept in a safe and usable condition
KU27. personal protective equipment (PPE) and clothing that must be worn during operational activity and from where can it be obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document call logs, reports, task lists, and schedules with co-workers
GS2. prepare status and progress reports
GS3. record customers discussions in the call logs
GS4. write memos and e-mail to customers, co-workers, and vendors to provide them with work updates and to request appropriate information without english language errors regarding grammar or sentence construct
GS5. read about new products and services with reference to the organization and also from external forums such as websites and blogs
GS6. keep abreast with the latest knowledge by reading brochures, pamphlets, and product information sheets
GS7. read comments, suggestions, and responses to frequently asked questions (faqs) posted on the helpdesk portal
GS8. discuss task lists, schedules, and work-loads with co-workers
GS9. question customers appropriately in order to understand the nature of the problem and make a diagnosis
GS10. give clear instructions to customers
GS11. keep customers informed about progress
GS12. avoid using jargon, slang or acronyms when communicating with a customer, unless it is required
GS13. undertake basic numerical computations and calculations numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
GS14. identify various basic, compound and solid shapes as per dimensions given
GS15. basic shapes: square, rectangle, triangle, circle, quadrilaterals
GS16. compound shapes: involving squares, rectangles, triangles, circles, semi- circles, quadrants of a circle
GS17. solid shapes: cube, rectangular prism, cylinder
GS18. use appropriate measuring techniques and units of measurement
GS19. use appropriate units and number systems to express degree of accuracy
GS20. units and number systems representing degree of accuracy: decimals places,
GS21. significant figures, fractions as a decimal quantity use metric systems of measurement
GS22. participate on-the-job and other learning, training and development interventions and assessments
GS23. clarify task related information with appropriate personnel or technical adviser
GS24. follow organization rule-based decision making process

GS25. take decision with systematic course of actions and/or response

GS26. understand importance of proper documentation

GS27. planning and organization of tasks to meet deadlines

GS28. build customer relationships and use customer centric approach

GS29. seek and comprehend operation related inputs for clarification

GS30. find ways of modifying difficult operating stages to make them operation friendly

GS31. apply domain information to set and define operation parameters that ensures economy and quality of the product

GS32. critically evaluate operation parameters in relation to product features intended develop holistic and comprehensive profile of products based on segregated

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working safely	**3**	**7**	**-**	**-**
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	1	2	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations	1	2	-	-
PC3. work following laid down procedures and instructions	1	1	-	-
PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location	-	1	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	-	1	-	-
Prepare for erection of Power Distribution Linea and Substation	**4**	**7**	**-**	**-**
PC6. identify job requirements for specific operations as per instructions given from valid sources valid sources: job instruction sheet/job card; work drawings; supervisor/in- charge	1	1	-	-
PC7. brief team members as per requirement, agree and clarify role and job requirements and specifications	1	1	-	-
PC8. ensure equipment and tools required for distribution installation work are identified, acquired, calibrated, suitable and approved for use	1	1	-	-
PC9. carry out survey of proposed route of line	-	2	-	-
PC10. identify, estimate and acquire correct materials required for the installation work	-	1	-	-
PC11. ensure loading and unloading operations for pole parts in a safe and efficient manner	1	1	-	-
Erection of Power Distribution Lines and Sub- Stations	**14**	**42**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. determine pole location(s) as per approved procedures	1	2	-	-
PC13. ensure excavation operations are carried out with the help of ground crew for pole setting template, as per requirement and specifications, in a safe and efficient manner	1	2	-	-
PC14. perform pole erection procedures as per requirements and specifications, in a safe and efficient manner	1	2	-	-
PC15. install grounding for pole installation where required and cross arm fixing to the pole before erection	1	1	-	-
PC16. ensure pit filling and concreting is done as per requirement, as correct procedures	1	3	-	-
PC17. follow applicable construction standards e.g. rec construction standards, for carrying out the erection procedures	1	2	-	-
PC18. install pole guys/ stays and anchors as required, as per standard procedure	1	1	-	-
PC19. perform stay wire assembly as per requirements and specifications, safely and efficiently	1	2	-	-
PC20. install travelers on poles or insulators	1	1	-	-
PC21. temporarily run conductor/rope through travelers to reduce friction when sagging	1	3	-	-
PC22. attach pulling equipment to conductor/rope	1	3	-	-
PC23. set up and operate stringing equipment when using tension stringing method, derrick method.	-	2	-	-
PC24. carry out conductor stringing procedures, paving conductor on the ground along the pole taking into account permissible span length and sagging	-	2	-	-
PC25. transfer conductor from travelers to insulators	-	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. secure conductor using clamps or ties	-	2	-	-
PC27. carry out erection of poles mounted distribution sub-station as per rec constructional standard f2	1	2	-	-
PC28. ensure maintenance of necessary clearances for execution of poles mounted distribution sub-station	1	3	-	-
PC29. carry out earthing arrangements for pole mounted distribution sub-station as per rec constructional standard f10	-	2	-	-
PC30. ensure 3 separate earthing with each for lightining arrestor, transformer body with metal parts and neutral	1	2	-	-
PC31. ensure proper rating of lt mains cable & fuses (ht & lt) during pole mounting sub-station installation	-	2	-	-
Commissioning	**3**	**14**	**-**	**-**
PC32. thoroughly check the line for clearances	1	2	-	-
PC33. check guarding and stays for correctness and suitability	-	2	-	-
PC34. install warning devices and signage	-	2	-	-
PC35. inspect the pole and related components to check if it is as per specification and without defects	-	2	-	-
PC36. ensure that before commissioning the distribution transformer, all earthing connections are carried out and earth resistance is maintained within limits	1	2	-	-
PC37. check breather, oil level in the conservator, ht & lt side jumper connections	1	2	-	-
PC38. carry out pre-commissioning test of distribution transformer viz. ir test, continuity test etc	-	2	-	-
Post Erection activities	**-**	**6**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC39. shut down and store equipment to a safe condition on completion of the activities	-	2	-	-
PC40. leave the work area in a safe and tidy condition on completion of the erection activities	-	2	-	-
PC41. refer unresolved job related problems to appropriate personnel for support	-	1	-	-
PC42. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	1	-	-
NOS Total	**24**	**76**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0106
NOS Name	Erection of Power Distribution Lines and Substations
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection & Commissioning
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022

PSS/N0107: Operation and Maintenance of an 11/0.433 KV Distribution Substation

Description

This unit covers the competencies required by lineman distribution to operate andmaintain 11/0.433 KV Distribution Substation. This includes working with the crew,handling of tools and equipment for operation & maintenance and carrying outnecessary tasks in a safe, efficient and effective manner.

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations

PC3. work following laid down procedures and instructions

PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location

PC5. ensure work area is clean and safe from hazards before and after the job is completed

PC6. inspect the component to check if it is as per specification and without defects

Operate and maintain 11/0.433 KV Distribution Substation

To be competent, the user/individual on the job must be able to:

PC7. identify job requirements for specific operations as per instructions given from valid sourcesvalid sources: job instruction sheet/job card; work drawings; supervisor/incharge

PC8. identify various components of the power system

PC9. ensure equipment and tools required for installation work are identified, acquired, calibrated, suitable and approved for use

PC10. identify, estimate and acquire correct materials required for the substation erection and installation work

PC11. follow standard specifications and procedures for installing a pole mounted distribution transformer

PC12. ensure poles set to proper depth, and properly aligned

PC13. carry out erection of channel on the double pole for preparation of transformer bed as per requirement

PC14. fix lightning arrester as per requirement and standard procedure

PC15. install earth connection as per standard procedure

PC16. install cross arm as per specifications and requirement

PC17. provide anti-climbing device on poles

PC18. arrange to lift the transformer and put it on the transformer bed in a safe and efficient manner

PC19. fit the gang operating (go switch) and dropout fuse as per standard procedure

PC20. follow applicable construction standards e.g. rec construction standards, for carrying out the erection procedures

PC21. connect low voltage cables as per standard procedures in a safe and efficient manner

PC22. carry out low voltage able joints as per standard procedures, safely and effectively

PC23. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area

PC24. check oil level and ensure leakages are attended to and arrested

PC25. check oil bdv and acidity at regular intervals as per schedule and standard procedure

PC26. checking for sludge, dust, dirt ,moisture ion in oil and address it effectively in a timely fashion

PC27. clean bushings regularly and inspect for any cracks

PC28. check, note and rectify dust & dirt deposition, salt or chemical deposition, cement or acid fumes depositions

PC29. check tap position and gap of arching horn and tighten connection as requirement to address any issues

PC30. check neutral grounding and ensure it is maintained as per standard

PC31. periodically check for any loose connections of the terminations of hv & lv side

PC32. examine the breather through color of silica gel , if pink heat it or replace if necessary

Post erection activities

To be competent, the user/individual on the job must be able to:

PC33. ensure facility is locked and warning signs are displayed effectively

PC34. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved

PC35. leave the work area in a safe and tidy condition on completion of the substation construction and maintenance activities

PC36. refer unresolved job related problems to appropriate personnel for support

PC37. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. relevant health and safety requirements applicable in the work place

KU3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. how to engage with specialists for support in order to resolve incidents and service requests

KU6. importance of working in clean and safe environment practices and procedures

KU7. relevant people and their responsibilities within the work area

KU8. escalation matrix and procedures for reporting work and employment related issues

KU9. various components of the power system Components: e.g. transformers, Isolators, CTs, PTs, Circuit breakers, LAs, various types of Panels & Sub-station protection systems

KU10. various components of the power system Components: e.g. transformers, Isolators, CTs, PTs, Circuit breakers, LAs, various types of Panels & Sub-station protection systems

KU11. specific health and safety precautions which must be taken when carrying out substation installation processes

KU12. hazards associated with carrying out substation construction and installation process and maintenance, and how they can be minimized Hazards: e.g. live wires and equipment, heavy objects, insects and reptiles, obstructions and blockages, sharp edges and equipment, etc

KU13. importance of following job instructions and defined installation and maintenance procedures

KU14. equipment used in substation construction and maintenance activities

KU15. importance of leaving the work area and equipment in a safe and clean condition on completion of the heat treatment activities

KU16. importance of reporting problems in a timely manner

KU17. methods and parameters to check quality of the components against required quality standards

KU18. types of cable joints Types: e.g. straight, T-joint, terminal joint

KU19. calibration schedule of all equipment used in the construction and maintenance procedures

KU20. importance of tools and equipment to be kept in a safe and usable condition

KU21. importance of displaying rating and diagram plates

KU22. personal protective equipment (ppe) and clothing that must be worn during the heat treatment activity and from where can it be obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in english and/or local language

GS2. fill up appropriate technical forms, process charts, activity logs as per organizational format in english and/or local language

GS3. convey and share technical information clearly using appropriate language

GS4. check and clarify task-related information

GS5. liaise with appropriate authorities using correct protocol

GS6. communicate with people in respectful form and manner in line with organizational protocol

GS7. undertake basic numerical computations and calculations numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages

GS8. identify various basic, compound and solid shapes as per dimensions given basic shapes: square, rectangle, triangle, circle, quadrilaterals compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle solid shapes: cube, rectangular prism, cylinder

GS9. use appropriate measuring techniques and units of measurement

GS10. use appropriate units and number systems to express degree of accuracy units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity

GS11. use metric systems of measurement

GS12. participate in on-the-job and other learning, training and development interventions and assessments

GS13. clarify task related information with appropriate personnel or technical adviser

GS14. seek to improve and modify own work practices

GS15. maintain current knowledge of application standards, legislation, codes of practice and product/process developments

GS16. identify problems with work planning, procedures, output and behavior and their implications

GS17. prioritize and plan for problem solving

GS18. communicate problems appropriately to others

GS19. identify sources of information and support for problem solving

GS20. seek assistance and support from other sources to solve problems

GS21. identify effective resolution techniques

GS22. select and apply resolution techniques

GS23. seek evidence for problem resolution

GS24. plan, prioritize and sequence work operations as per job requirements

GS25. organize and analyze information relevant to work

GS26. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time

GS27. undertake and express new ideas and initiatives to others

GS28. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses

GS29. participate in improvement procedures including process, quality and internal/external customer/supplier relationships

GS30. ones competencies in new and different situations and contexts to achieve more

GS31. exercise restraint while expressing dissent and during conflict situations

GS32. avoid and manage distractions to be disciplined at work

GS33. manage own time for achieving better results

GS34. work in a team in order to achieve better results

GS35. identify and clarify work roles within a team

GS36. communicate and cooperate with others in the team for better results

GS37. seek assistance from fellow team members

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working safely	**4**	**11**	**-**	**-**
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	1	2	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations	1	2	-	-
PC3. work following laid down procedures and instructions	1	1	-	-
PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location	-	2	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	-	2	-	-
PC6. inspect the component to check if it is as per specification and without defects	1	2	-	-
Operate and maintain 11/0.433 KV Distribution Substation	**19**	**55**	**-**	**-**
PC7. identify job requirements for specific operations as per instructions given from valid sourcesvalid sources: job instruction sheet/job card; work drawings; supervisor/incharge	1	2	-	-
PC8. identify various components of the power system	1	1	-	-
PC9. ensure equipment and tools required for installation work are identified, acquired, calibrated, suitable and approved for use	-	2	-	-
PC10. identify, estimate and acquire correct materials required for the substation erection and installation work	-	2	-	-
PC11. follow standard specifications and procedures for installing a pole mounted distribution transformer	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure poles set to proper depth, and properly aligned	-	2	-	-
PC13. carry out erection of channel on the double pole for preparation of transformer bed as per requirement	2	3	-	-
PC14. fix lightning arrester as per requirement and standard procedure	2	2	-	-
PC15. install earth connection as per standard procedure	1	2	-	-
PC16. install cross arm as per specifications and requirement	1	2	-	-
PC17. provide anti-climbing device on poles	-	2	-	-
PC18. arrange to lift the transformer and put it on the transformer bed in a safe and efficient manner	-	3	-	-
PC19. fit the gang operating (go switch) and dropout fuse as per standard procedure	2	3	-	-
PC20. follow applicable construction standards e.g. rec construction standards, for carrying out the erection procedures	2	2	-	-
PC21. connect low voltage cables as per standard procedures in a safe and efficient manner	1	2	-	-
PC22. carry out low voltage able joints as per standard procedures, safely and effectively	1	2	-	-
PC23. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area	-	2	-	-
PC24. check oil level and ensure leakages are attended to and arrested	-	2	-	-
PC25. check oil bdv and acidity at regular intervals as per schedule and standard procedure	1	2	-	-
PC26. checking for sludge, dust, dirt ,moisture ion in oil and address it effectively in a timely fashion	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. clean bushings regularly and inspect for any cracks	-	2	-	-
PC28. check, note and rectify dust & dirt deposition, salt or chemical deposition, cement or acid fumes depositions	-	2	-	-
PC29. check tap position and gap of arching horn and tighten connection as requirement to address any issues	1	2	-	-
PC30. check neutral grounding and ensure it is maintained as per standard	1	2	-	-
PC31. periodically check for any loose connections of the terminations of hv & lv side	-	2	-	-
PC32. examine the breather through color of silica gel , if pink heat it or replace if necessary	-	2	-	-
Post erection activities	**-**	**11**	**-**	**-**
PC33. ensure facility is locked and warning signs are displayed effectively	-	2	-	-
PC34. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	3	-	-
PC35. leave the work area in a safe and tidy condition on completion of the substation construction and maintenance activities	-	2	-	-
PC36. refer unresolved job related problems to appropriate personnel for support	-	2	-	-
PC37. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	2	-	-
NOS Total	**23**	**77**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0107
NOS Name	Operation and Maintenance of an 11/0.433 KV Distribution Substation
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	25/07/2017
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N0108: Laying of Underground and AB Cables

Description

This unit covers the competencies required technicians to lay underground and AB cables for setting up Power Distribution Lines. This includes working with the crew to dig trenches, prepare and lay wiring, handling of tools and equipment for laying andcommissioning and carrying out necessary tasks in a safe, efficient and effective manner.The candidate will be expected to perform mostly independently, under little to no supervision.

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations

PC3. work following laid down procedures and instructions

PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location

PC5. ensure work area is clean and safe from hazards before and after the job is completed

Preparing cables and other materials for job

To be competent, the user/individual on the job must be able to:

PC6. identify job requirements for specific operations as per instructions given from valid sourcesValid sources: job instruction sheet/job card; work drawings; supervisor/in- charge

PC7. brief team members as per requirement, agree and clarify role and job requirements and specifications

PC8. ensure all tools, equipment and material supplies required for the work are acquired and transported safely to the work site

PC9. check tools and equipment for calibration and assess suitability for use

PC10. check and select the correct types of cables for the job

PC11. ensure the cable and joints are suitable and as per job requirement

Laying of cables

To be competent, the user/individual on the job must be able to:

PC12. determine cable installation and laying location(s) as per approved procedures

PC13. ensure the trench digging operations have been completed as per requirement and specifications, in a safe and efficient manner

PC14. lay down cable as per requirement, including cleaning, lubricating, setting of conduit and pulling cables through conduit safely and without damage

PC15. pull cable through conduit using equipment such as tension machines, winches and capstans

PC16. ensure cables are set to proper depth, and properly aligned

PC17. replace cables where not as per requirement

PC18. ensure pit back filling, brick laying and concreting is done as per requirement, as correct procedures

PC19. follow applicable construction standards e.g. rec construction standards, for carrying out the laying procedures

PC20. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area

Carrying out maintenance

To be competent, the user/individual on the job must be able to:

PC21. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved

PC22. leave the work area in a safe and tidy condition on completion of the laying activities

PC23. refer unresolved job related problems to appropriate personnel for support

PC24. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. relevant health and safety requirements applicable in the work place

KU3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. how to engage with specialists for support in order to resolve incidents and service requests

KU6. importance of working in clean and safe environment practices and procedures

KU7. relevant people and their responsibilities within the work area

KU8. escalation matrix and procedures for reporting work and employment related issues

KU9. types of common cables and their use types: core (one, three); armored and unarmored; LT & HT; use: e.g. single and three phase systems, etc.

KU10. common electric and power terminology used in transmission and distribution

KU11. different types of insulation used in cables and their purpose types : e.g. rubber, paper, PVC, XLPE

KU12. conductor metal types, composition and shapes types: e.g. copper, aluminum

KU13. importance and classification of cable with respect to insulation thickness specific health and safety precautions which must be taken when carrying out cable laying processes and working in confined spacesprecautions: e.g. loose dhotis, pajamas, key chain or watch chains should not be worn; shoes with projecting nails or other types of metal parts not to be used; do not start work unless circuit is in off condition, line clear permit is aken on equipment, equipment or line is properly earthed, every electrical line or equipment should be first made off and take line clear permit before taking the work in hand

KU14. hazards associated with carrying out cable laying processes and how they can be minimized hazards: live wires and equipment, blockages and obstructions, loose earth, sharp surfaces and edges, insects and reptiles, heavy objects, etc.

KU15. importance of following job instructions and defined cable laying procedures

KU16. material preparation methods and techniques to be undertaken, prior to laying cables

KU17. tools and equipment used in cable laying activities

KU18. preparation of cables and equipment for cable laying activities

KU19. types of cable joints types: e.g. straight, t-joint,

KU20. types of conduit systems and components

KU21. adjacent utilities such as gas, water, communication and drainage requirements

KU22. pulling methods and calculations

KU23. installation specifications such as direct burial and duct system

KU24. voltage and amperage

KU25. problems that can occur with the cable laying and maintenance operations, and how these can be overcome

KU26. AB cables, their types, stringing and jointing methods

KU27. procedures for handling components with imperfections/defects that cannot be removed/repaired and how can they be minimized

KU28. importance of leaving the work area and equipment in a safe and clean condition on completion of the job activities

KU29. importance of reporting problems in a timely manner

KU30. calibration schedule of all equipment used in heat treatment procedure

KU31. keep records of the job including data logging, chart recording of various activities and data points like tolerance levels, etc.

KU32. importance of tools and equipment to be kept in a safe and usable condition

KU33. personal protective equipment (PPE) and clothing that must be worn during the cable laying and maintenance activity and from where can it be obtained ppe: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), zola, safety rope

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document call logs, reports, task lists, and schedules with co-workers

GS2. prepare status and progress reports

GS3. record customers discussions in the call logs

GS4. write memos and e-mail to customers, co-workers, and vendors to provide them with work updates and to request appropriate information without english language errors regarding grammar or sentence construct

GS5. read about new products and services with reference to the organization and also from external forums such as websites and blogs

GS6. keep abreast with the latest knowledge by reading brochures, pamphlets, and product information sheets

GS7. read comments, suggestions, and responses to frequently asked questions (faqs) posted on the helpdesk portal

GS8. discuss task lists, schedules, and work-loads with co-workers

GS9. question customers appropriately in order to understand the nature of the problem and make a diagnosis

GS10. give clear instructions to customers

GS11. keep customers informed about progress

GS12. avoid using jargon, slang or acronyms when communicating with a customer, unless it is required

GS13. undertake basic numerical computations and calculations

GS14. numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages

GS15. identify various basic, compound and solid shapes as per dimensions given

GS16. basic shapes: square, rectangle, triangle, circle, quadrilaterals

GS17. compound shapes: involving squares, rectangles, triangles, circles, semi- circles, quadrants of a circle

GS18. solid shapes: cube, rectangular prism, cylinder

GS19. use appropriate measuring techniques and units of measurement

GS20. use appropriate units and number systems to express degree of accuracy

GS21. units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity use metric systems of measurement

GS22. participate in on-the-job and other learning, training and development interventions and assessments

GS23. clarify task related information with appropriate personnel or technical adviser

GS24. seek to improve and modify own work practices maintain current knowledge of application standards, legislation, codes of practice and product/process developments

GS25. follow organization rule-based decision making process

GS26. take decision with systematic course of actions and/or response

GS27. understand importance of proper documentation

GS28. planning and organization of tasks to meet deadlines

GS29. build customer relationships and use customer centric approach

GS30. seek and comprehend operation related inputs for clarification

GS31. find ways of modifying difficult operating stages to make them operation friendly

GS32. apply domain information to set and define operation parameters that ensures economy and quality of the product

GS33. critically evaluate operation parameters in relation to product features intended develop holistic and comprehensive profile of products based on segregated

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working safely	**3**	**15**	**-**	**-**
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	1	2	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations	1	4	-	-
PC3. work following laid down procedures and instructions	1	3	-	-
PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location	-	3	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	-	3	-	-
Preparing cables and other materials for job	**6**	**18**	**-**	**-**
PC6. identify job requirements for specific operations as per instructions given from valid sourcesValid sources: job instruction sheet/job card; work drawings; supervisor/in- charge	2	3	-	-
PC7. brief team members as per requirement, agree and clarify role and job requirements and specifications	1	3	-	-
PC8. ensure all tools, equipment and material supplies required for the work are acquired and transported safely to the work site	1	3	-	-
PC9. check tools and equipment for calibration and assess suitability for use	-	3	-	-
PC10. check and select the correct types of cables for the job	1	3	-	-
PC11. ensure the cable and joints are suitable and as per job requirement	1	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Laying of cables	**15**	**31**	**-**	**-**
PC12. determine cable installation and laying location(s) as per approved procedures	2	5	-	-
PC13. ensure the trench digging operations have been completed as per requirement and specifications, in a safe and efficient manner	2	3	-	-
PC14. lay down cable as per requirement, including cleaning, lubricating, setting of conduit and pulling cables through conduit safely and without damage	2	5	-	-
PC15. pull cable through conduit using equipment such as tension machines, winches and capstans	1	3	-	-
PC16. ensure cables are set to proper depth, and properly aligned	1	3	-	-
PC17. replace cables where not as per requirement	2	3	-	-
PC18. ensure pit back filling, brick laying and concreting is done as per requirement, as correct procedures	2	3	-	-
PC19. follow applicable construction standards e.g. rec construction standards, for carrying out the laying procedures	2	3	-	-
PC20. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area	1	3	-	-
Carrying out maintenance	**-**	**12**	**-**	**-**
PC21. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	3	-	-
PC22. leave the work area in a safe and tidy condition on completion of the laying activities	-	3	-	-
PC23. refer unresolved job related problems to appropriate personnel for support	-	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	3	-	-
NOS Total	**24**	**76**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0108
NOS Name	Laying of Underground and AB Cables
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection & Commissioning
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette,discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required

PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt

PC3. give information to others clearly, at a pace and in a manner that helps them to understand

PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible

PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks

PC6. display appropriate communication etiquette while working

PC7. display active listening skills while interacting with others at work

PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

PC9. demonstrate responsible and disciplined behaviors at the workplace

PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. eescalation matrix and procedures for reporting work and employment related issues

KU5. various categories of people that one is required to communicate and co-ordinate with in the organization

KU6. importance of effective communication in the workplace

KU7. importance of teamwork in organizational and individual success

KU8. various components of effective communication

KU9. key elements of active listening

KU10. value and importance of active listening and assertive communication

KU11. barriers to effective communication
KU12. importance of tone and pitch in effective communication
KU13. importance of avoiding casual expletives and unpleasant terms while communicating professional circles
KU14. how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
KU15. importance of ethics for professional success
KU16. importance of discipline for professional success
KU17. what constitutes disciplined behavior for a working professional
KU18. common reasons for interpersonal conflict
KU19. importance of developing effective working relationships for professional success
KU20. expressing and addressing grievances appropriately and effectively
KU21. importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note the information communicated by the officer incharge
GS2. note down observations (if any) related to the operation/maintenance
GS3. read and interpret the process required for different types of manuals
GS4. read and interpret the flowchart of all parts of an assembly
GS5. read manuals and documents to understand the product-details & how they can be used
GS6. discuss task lists, schedules and activities with the colleague/supervisor
GS7. effectively communicate with the team members
GS8. attentively listen and comprehend the information given by the colleague/supervisor/contractor
GS9. communicate clearly with the colleague on the issues faced during query/fault
GS10. follow colleague/contractor rule-based decision making process
GS11. take decisions with systematic course of actions and/or response
GS12. planning and organization of tasks to meet deadlines
GS13. build customer relationships and use customer centric approach
GS14. seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
GS15. work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
GS16. critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
working with others	**30**	**70**	**-**	**-**
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

PC1. use protective clothing/equipment for specific tasks and work conditions
PC2. state the name and location of people responsible for health and safety in the workplace
PC3. state the names and location of documents that refer to health and safety in the workplace
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems
PC7. use standard safe working practices when working at heights, confined areas and trenches
PC8. test any electrical equipment and system using insulated testing devices before touching them
PC9. ensure positive isolation of electrical equipment & system as per given standards
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others
PC12. state methods of accident prevention in the work environment of the job role
PC13. state location of general health and safety equipment in the workplace
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage
PC17. store flammable materials and machine lubricating oil safely and correctly
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards
PC19. ensure proper working condition of battery and battery charger
PC20. maintain electrolyte level of each cell with distilled water
PC21. maintain proper ventilation in battery room
PC22. apply good housekeeping practices at all times
PC23. identify common hazard signs displayed in various areas

PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible

KU8. possible causes of risk and accident
KU9. methods of accident prevention
KU10. safe working practices when working with tools and machines
KU11. safe working practices while working at various hazardous sites
KU12. where to find all the general health and safety equipment in the workplace
KU13. various dangers associated with the use of electrical equipment
KU14. positive isolation of electrical equipment and system
KU15. safe handling and disposal of hazardous power plant wastes
KU16. use of emission and pollution control devices and measures taken to control pollution
KU17. various safety procedures and equipment used to work at heights, trenches and confined places
KU18. safe working practices specific to working with electrical equipment & system e.g. lock out/ tag out, ptw, etc.
KU19. preventative and remedial actions to be taken in the case of exposure to toxic materials
KU20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
KU21. precautionary activities taken to prevent fire accident
KU22. various causes of fire
KU23. techniques of using the different fire extinguishers
KU24. different methods of extinguishing fire
KU25. different materials used for extinguishing fire
KU26. emergency rescue techniques applied during a fire hazard
KU27. various types of safety signs and what they mean
KU28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note the information communicated by the officer incharge.
GS2. note down observations (if any) related to the operation/maintenance.
GS3. read and interpret the process required for different types of manuals for maintenance.
GS4. read and interpret the flowchart of all parts of an assembly.
GS5. read manuals and documents to understand the product-details & how they can be used.
GS6. discuss task lists, schedules and activities with the colleague/supervisor.
GS7. effectively communicate with the team members.
GS8. attentively listen and comprehend the information given by the colleague/supervisor/contractor.
GS9. communicate clearly with the colleague on the issues faced during query/fault.
GS10. follow colleague/contractor rule-based decision making process.
GS11. take decisions with systematic course of actions and/or response.

GS12. planning and organization of tasks to meet deadlines.

GS13. build customer relationships and use customer centric approach.

GS14. seek and comprehend operation related inputs for clarification

GS15. find ways of modifying difficult operating stages to make it operation friendly

GS16. work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.

GS17. quick approach and solution towards faults repairing.

GS18. critically evaluate operation parameters in relation to system normality

GS19. develop a holistic and comprehensive profile of grid station on segregated discrete process.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Health and safety	**11**	**46**	**-**	**-**
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	**4**	**9**	**-**	**-**
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
Emergencies, rescue and first-aid procedures	**9**	**21**	**-**	**-**
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	**24**	**76**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N0105.Repair and Maintenance of Power Distribution Lines and components	25	75	0	0	100	20
PSS/N0106.Erection of Power Distribution Lines and Substations	24	76	-	-	100	20
PSS/N0107.Operation and Maintenance of an 11/0.433 KV Distribution Substation	23	77	-	-	100	20
PSS/N0108.Laying of Underground and AB Cables	24	76	-	-	100	20
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
Total	**150**	**450**	**0**	**0**	**600**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
KWH	Kilo Watt Hour

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.